

Quantum Superposition Decay for Quantum Complexity Classes

A Classical–Quantum Bridge via the Read-Write-Execute Memory Model

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June 23, 2026

Abstract

We introduce *Superposition Decay* (SD) as a formal measurement window that governs the collapse of a quantum superposition state on a classical computing substrate. Building on standard quantum postulates and the wave-function collapse principle, we show that any classical algorithm bounded by linear time complexity—deterministic or non-deterministic—can be evaluated in parallel via superposition decay on a quantum system. We further demonstrate that all problems belonging to a system of linear equations can be expressed as a probabilistic directed acyclic graph (DAG) on such a system. A concrete Read-Write-Execute (RWX) memory model formalises the decay window in operational terms: four write operations and two read operations reduce atomically to a single coherent read. We map these results onto the standard quantum complexity hierarchy ($P \subseteq BPP \subseteq BQP \subseteq PP \subseteq PSPACE$) and characterise which promise-problem classes are tractable under superposition decay. The framework is then extended in two directions: (i) **Rift**, a quantum-classical grammar matcher whose parser exploits grammar superposition to evaluate all production rules in parallel, collapsing to a unique parse tree at a grammar decay event; and (ii) the **Quantum Burst Protocol (QBP)**, a communication protocol in which all candidate network paths are held in superposition during handshake and selected by a single measurement event. Both components are implemented within the MMUKO OS / OBINexus toolchain via the pipeline `riftlang.exe → .so.a → rift.exe → gosilang`, orchestrated by `nlink` and `polybuild`.

Contents

1	Introduction	2
2	Preliminaries	2
2.1	Quantum Postulates	2
2.2	Classical Turing Machine and Complexity	3
2.3	Quantum Circuits	3
2.4	Probabilistic Directed Acyclic Graphs	3
3	Superposition Decay	3
3.1	Informal Motivation	3
3.2	Formal Definition	4
3.3	Constructive and Destructive Superposition	4
4	The Read-Write-Execute (RWX) Memory Model	5
4.1	Motivation	5
4.2	Formal Model	5
4.3	RWX Permission Table	5
5	Complexity Classes Under Superposition Decay	6
5.1	Promise Problem Formulation	6
5.2	Complexity Class Definitions	6
5.3	Tractability Under SD	6
6	The Main Parallelism Theorem	7
6.1	Statement	7
6.2	Linear Systems as Parallel DAGs	8
7	Rift: A Quantum-Classical Grammar Matcher	8
7.1	Overview	8
7.2	Wave Postulates as Triangular Matching Postulates	8
7.3	Parallel Grammar Evaluation	9
7.4	Stabilisation Principle	9
8	Quantum Burst Protocol (QBP)	9
8.1	Motivation	9
8.2	Formal Definition	9
8.3	QBP Handshake as Superposition Decay	10
8.4	Spectral Graph Representation	10
9	MMUKO OS and the OBINexus Toolchain	11
9.1	MMUKO OS	11
9.2	Toolchain Stack	11
10	Conclusion and Future Work	12
10.1	Future Work	12

1 Introduction

The relationship between quantum and classical computation is most precisely characterised through the lens of promise-problem complexity classes. While quantum speedup for specific problem families (integer factorisation, unstructured search) is well established, a general framework that maps the *measurement event*—the moment a quantum superposition collapses to a definite classical outcome—onto classical computational primitives has received less systematic attention.

This paper proposes such a framework under the name **Superposition Decay** (SD). The central intuition is that the collapse of a superposition is not a passive observation but an active computational window: a bounded interval of control during which a system may coherently commit or roll back parallel computations, analogous to a transactional memory operation.

We formalise five related contributions:

- C1. The SD window.** A formal definition of the measurement window as a triple (resolved / rejected / pending), bounding acceptance and rejection by probability thresholds $a(n)$ and $b(n)$, with an intermediate *pending* region supporting deferred evaluation.
- C2. The RWX memory model.** A Read-Write-Execute abstraction that maps quantum measurement to classical memory transactions: n writes followed by k reads collapse to a single coherent read event when the last read is the terminal operation.
- C3. Complexity-class tractability under SD.** A characterisation showing that BQP promise problems are exactly the class tractable under a single superposition decay event; problems in PP and PSPACE require multiple decay events or additional measurement principles.
- C4. Rift: quantum-classical grammar matching.** A formalisation of the Rift compiler’s parser as a quantum automaton, with grammar production rules as superposed branches and parse commitment as a grammar decay event governed by the stabilisation principle.
- C5. Quantum Burst Protocol (QBP).** A communication protocol in which all candidate network paths are held in superposition during connection negotiation and selected by a single measurement (spectral decay), achieving optimal routing in $O(1)$ measurement time.

The paper is organised as follows. Section 2 reviews necessary background in quantum mechanics and complexity theory. Section 3 defines superposition decay formally. Section 4 presents the RWX memory model. Section 5 maps results onto the complexity hierarchy. Section 6 states and proves the main parallelism theorem. Section 7 formalises Rift as a quantum-classical grammar matcher. Section 8 introduces QBP. Section 9 describes the MMUKO OS and OBINexus toolchain. Section 10 concludes.

2 Preliminaries

2.1 Quantum Postulates

We work in the standard Hilbert-space formulation of quantum mechanics. A quantum state is represented by a unit vector $|\psi\rangle \in \mathbb{C}^{2^n}$ for an n -qubit system.

Axiom 2.1 (State Postulate). The complete state of a quantum system at time t is described by a normalised vector $|\psi(t)\rangle \in \mathcal{H}$, the system’s Hilbert space, satisfying $\langle\psi(t) | \psi(t)\rangle = 1$.

Axiom 2.2 (Superposition Postulate). If $|\psi_1\rangle$ and $|\psi_2\rangle$ are valid states of a quantum system, then any linear combination

$$|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle, \quad c_1, c_2 \in \mathbb{C}, \quad |c_1|^2 + |c_2|^2 = 1,$$

is also a valid state. This follows from the linearity and homogeneity of the Schrödinger equation.

Axiom 2.3 (Wave-Function Collapse / Measurement Postulate). Upon measurement of an observable $\hat{O}$ with eigenvalues $\{\lambda_i\}$ and eigenstates $\{|e_i\rangle\}$, the system transitions from $|\psi\rangle = \sum_i c_i |e_i\rangle$ to eigenstate $|e_j\rangle$ with probability $|c_j|^2$. The post-measurement state is $|e_j\rangle$.

Axiom 2.4 (Evolution Postulate). Between measurements, a closed quantum system evolves unitarily: $|\psi(t)\rangle = U(t) |\psi(0)\rangle$, where $U(t)$ is a unitary operator.

2.2 Classical Turing Machine and Complexity

A *Turing machine* (TM) is a tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$ with the standard interpretation. Time complexity $T(n)$ denotes the maximum number of steps on inputs of length n . We use the standard complexity classes P, NP, BPP, PP, PSPACE, and BQP; their promise-problem definitions are recalled in Section 5.

2.3 Quantum Circuits

A *quantum circuit* Q_n on n qubits is a sequence of gates drawn from a universal gate set (e.g., $\{H, \text{CNOT}, T\}$) that accepts an n -qubit input and produces a 1-qubit output. A *uniform circuit family* $\mathcal{Q} = \{Q_n : n \in \mathbb{N}\}$ is one where the description of Q_n is computable in polynomial time from n .

2.4 Probabilistic Directed Acyclic Graphs

Definition 2.5 (Probabilistic DAG). A *probabilistic directed acyclic graph* $G = (V, E, p)$ is a DAG where each edge $(u, v) \in E$ carries a probability weight $p(u, v) \in [0, 1]$ such that for every vertex u , $\sum_{v:(u,v) \in E} p(u, v) \leq 1$. Each path from a source to a sink represents a computation branch; the probability of a path is the product of its edge weights.

3 Superposition Decay

3.1 Informal Motivation

When a quantum system in superposition is measured, the resulting classical outcome is not predetermined—it is drawn from a probability distribution defined by the amplitudes. We argue that this *measurement event* is best understood not as an instantaneous point but as a *window*: a bounded region of the computation during which the system retains coherence and the outcomes of parallel branches may still be combined or discarded.

Once the window closes—the superposition *decays*—a single classical outcome is committed. The SD framework captures this window operationally.

3.2 Formal Definition

Definition 3.1 (Quantum Connection Graph). A *quantum connection* is a graph $G = (E, N)$ where N is the set of n -qubit nodes (quantum registers) and $E \subseteq N \times N$ is the set of entanglement edges. A path from node A to node B in G represents a sequence of entangled operations connecting two computational loci.

Definition 3.2 (Superposition State Vector). For an n -qubit system, the superposition state is

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \alpha_x \in \mathbb{C}, \quad \sum_x |\alpha_x|^2 = 1.$$

The probability of observing basis state $|x\rangle$ is $p(x) = |\alpha_x|^2$.

Definition 3.3 (Superposition Decay Window). Let $\mathcal{Q} = \{Q_n\}$ be a uniform quantum circuit family and let $a, b : \mathbb{N} \rightarrow [0, 1]$ be functions with $a(n) > b(n)$ for all n . The *superposition decay window* $\text{SD}(a, b)$ is the triple

$$\text{SD}(a, b) = (\text{resolved}, \text{rejected}, \text{pending}),$$

where for an input x of length n :

- x is **resolved** (accepted) if $\Pr[Q_{|x|}(x) = 1] \geq a(|x|)$,
- x is **rejected** if $\Pr[Q_{|x|}(x) = 1] \leq b(|x|)$,
- x is **pending** if $b(|x|) < \Pr[Q_{|x|}(x) = 1] < a(|x|)$.

The *decay event* is the transition of a state from *pending* to either *resolved* or *rejected* upon measurement.

Remark 3.4. The three-valued output (resolved / rejected / pending) generalises the standard binary outcome of a classical TM. The *pending* state corresponds precisely to the undecided region of a promise problem: the system is contractually guaranteed not to receive inputs from this region in the promise-problem formulation, but the SD framework makes the region explicit as a computational state.

3.3 Constructive and Destructive Superposition

The superposition principle admits both *constructive* interference (amplitudes add, increasing probability of a desired outcome) and *destructive* interference (amplitudes cancel, suppressing undesired outcomes). We formalise this in terms of the wave function as follows.

Definition 3.5 (Constructive / Destructive Interference). Given two states $|\psi_1\rangle = \sum_x \alpha_x |x\rangle$ and $|\psi_2\rangle = \sum_x \beta_x |x\rangle$, their superposition $|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ exhibits:

- **Constructive interference** at x if $|c_1 \alpha_x + c_2 \beta_x| > \max(|c_1 \alpha_x|, |c_2 \beta_x|)$.
- **Destructive interference** at x if $|c_1 \alpha_x + c_2 \beta_x| < \min(|c_1 \alpha_x|, |c_2 \beta_x|)$.

In the SD framework, a quantum algorithm exploits constructive interference to amplify the amplitude of the correct solution branch before the decay window closes, and destructive interference to suppress incorrect branches.

4 The Read-Write-Execute (RWX) Memory Model

4.1 Motivation

Classical memory access is characterised by read (R), write (W), and execute (X) permissions, typically represented as a triple (r, w, x) per memory region. We propose a quantum analogue in which the *sequence* of reads and writes within a superposition decay window determines the atomicity and coherence of the resulting classical observation.

4.2 Formal Model

Definition 4.1 (RWX Operation Sequence). An *RWX sequence* is a finite ordered list of operations $\sigma = (o_1, o_2, \dots, o_k)$ where each $o_i \in \{R, W, X\}$. The sequence is *terminal-read* if $o_k = R$ (the final operation is a read).

Definition 4.2 (Atomic Decay Unit). A terminal-read RWX sequence of w writes followed by r reads,

$$\sigma = (\underbrace{W, W, \dots, W}_w, \underbrace{R, R, \dots, R}_{r-1}, R),$$

constitutes a single *atomic decay unit* (ADU) equivalent to one coherent read operation. Formally:

$$w \cdot W + r \cdot R \xrightarrow{\text{SD}} 1 \cdot R_{\text{coherent}}.$$

Example 4.3 (4-Write 2-Read ADU). The concrete instance stated in the source document is:

$$4W + 2R \xrightarrow{\text{SD}} 1R.$$

This means: four parallel write operations (corresponding to four parallel computation branches in superposition) and two intermediate read probes collapse—upon superposition decay—into a single coherent classical read outcome. The dual-state nature of the system (write = superposed, read = collapsed) ensures that the terminal read is the only externally observable event.

Proposition 4.4 (Dual-State Equivalence). *In a quantum system respecting the SD window, the state before the terminal read is a superposition of all write branches. The terminal read collapses this superposition to a single classical value. Therefore, the external observer sees exactly one read operation regardless of the number of internal write branches.*

Proof. By the Measurement Postulate (Axiom 3), measurement of the system in state $|\Psi\rangle = \sum_x \alpha_x |x\rangle$ produces outcome x with probability $|\alpha_x|^2$ and collapses the state to $|x\rangle$. Each write branch corresponds to a basis state $|x\rangle$. Since the terminal read is the sole measurement, all prior writes are unobserved intermediate computations within the unitary evolution $U(t)$. By the Evolution Postulate (Axiom 4), $U(t)$ is reversible and does not produce classical outputs; only the terminal measurement does. Hence the external observable record contains exactly one read event. $\square$ $\square$

4.3 RWX Permission Table

The classical RWX permission system maps directly onto the decay window:

Permission	Classical Meaning	Quantum / SD Analogue
R (Read)	Load value from memory	Terminal measurement; collapses superposition
W (Write)	Store value to memory	Unitary gate application; maintains superposition
X (Execute)	Run code in memory region	Quantum circuit evaluation within decay window

5 Complexity Classes Under Superposition Decay

5.1 Promise Problem Formulation

A *promise problem* is a pair $(A_{\text{yes}}, A_{\text{no}})$ of disjoint languages. An algorithm *solves* the promise problem if it accepts every $x \in A_{\text{yes}}$ and rejects every $x \in A_{\text{no}}$, with no constraint on inputs outside $A_{\text{yes}} \cup A_{\text{no}}$.

5.2 Complexity Class Definitions

Class	Criteria
P	Promise problems for which a polynomial-time <i>deterministic</i> TM accepts all strings in A_{yes} and rejects all strings in A_{no} .
BPP	Promise problems for which a polynomial-time <i>probabilistic</i> TM accepts every string in A_{yes} with probability $\geq \frac{2}{3}$ and every string in A_{no} with probability $\leq \frac{1}{3}$.
BQP	Promise problems such that for functions $a, b : \mathbb{N} \rightarrow [0, 1]$ with $a(n) \geq \frac{2}{3}$ and $b(n) \leq \frac{1}{3}$, there exists a uniform quantum circuit family $\mathcal{Q} = \{Q_n : n \in \mathbb{N}\}$ where Q_n accepts n qubits and gives a 1-qubit output, such that every $x \in A_{\text{yes}}$ is accepted with probability $\geq a(x)$ and every $x \in A_{\text{no}}$ is accepted with probability $\leq b(x)$.
PP	Promise problems for which a polynomial-time probabilistic TM accepts every string in A_{yes} with probability $> \frac{1}{2}$ and every string in A_{no} with probability $\leq \frac{1}{2}$.
PSPACE	Promise problems for which a <i>deterministic</i> TM running in polynomial <i>space</i> accepts all A_{yes} and rejects all A_{no} .

Table 1: Complexity class definitions in the promise-problem framework.

5.3 Tractability Under SD

Theorem 5.1 (SD-Tractability of BQP). *A promise problem $(A_{\text{yes}}, A_{\text{no}})$ is in BQP if and only if it is solvable by a uniform quantum circuit family within a single superposition decay window $\text{SD}(a, b)$ with $a(n) - b(n) \geq \frac{1}{3}$ for all n .*

Proof. ($\Rightarrow$) By definition of BQP, there exists a uniform circuit family $\{Q_n\}$ with the stated acceptance probabilities. The circuit Q_n operates entirely within a unitary evolution (all gates are write operations in RWX terms) and terminates with a single measurement (the terminal read). This constitutes a single ADU as in Definition 4.2, hence falls within one SD window.

($\Leftarrow$) Suppose a single $\text{SD}(a, b)$ window with $a(n) - b(n) \geq \frac{1}{3}$ solves the promise problem. The circuit within the window is unitary (by the Evolution Postulate) and polynomial-time (by the polynomial-size constraint on the circuit family). This is exactly the definition of a BQP circuit. $\square$ $\square$

Corollary 5.2. $P \subseteq BPP \subseteq BQP$ under the SD framework, since deterministic and bounded-error probabilistic computations can be simulated by a single decay window with all amplitude concentrated on one branch.

Theorem 5.3 (Multi-Decay Requirement for PP). *There exist promise problems in $PP \setminus BQP$ that require more than one superposition decay event to resolve under SD.*

Proof Sketch. PP allows acceptance probabilities arbitrarily close to $\frac{1}{2}$, whereas BQP requires a gap of at least $\frac{1}{3}$. A problem instance where $\Pr[\text{accept}] = \frac{1}{2} + \epsilon$ for arbitrarily small $\epsilon > 0$ cannot be distinguished from a reject instance (with $\Pr = \frac{1}{2} - \epsilon$) by a single decay event with fixed thresholds a, b satisfying $a - b \geq \frac{1}{3}$. Multiple sequential decay events (with amplitude amplification between events) are required to drive the probability above the BQP threshold. $\square$ $\square$

Remark 5.4. PSPACE problems, being characterised by polynomial-space deterministic computation, lie outside the reach of a constant number of superposition decay events unless $BQP = PSPACE$, which is not currently believed. They require fundamentally different principles—either unbounded decay depth or oracle access—within the SD framework.

6 The Main Parallelism Theorem

6.1 Statement

Theorem 6.1 (Classical-to-Quantum Parallelism via SD). *Let $\mathcal{A}$ be a classical algorithm with time complexity $O(f(n))$ where f is a linear function ($f(n) = cn$ for some constant $c > 0$), and let $\mathcal{A}$ be deterministic or non-deterministic. Then there exists a quantum circuit family $\mathcal{Q}_{\mathcal{A}} = \{Q_n\}$ such that:*

1. Q_n simulates all execution branches of $\mathcal{A}$ on inputs of length n in parallel within a single superposition decay window $SD(a, b)$.
2. The circuit size of Q_n is polynomial in n .
3. The correct output is obtained upon the terminal measurement (the decay event) with probability $\geq \frac{2}{3}$.

Proof. We construct Q_n as follows.

Step 1: Encode branches. The k non-deterministic branches of $\mathcal{A}$ on input x correspond to basis states $|b_1\rangle, \dots, |b_k\rangle$ in a $\lceil \log_2 k \rceil$ -qubit register. Since $\mathcal{A}$ runs in linear time $O(n)$, the number of branches is at most $k \leq 2^{cn}$ for some constant c ; encoding requires $O(n)$ qubits.

Step 2: Superpose branches. Apply a Hadamard transform $H^{\otimes \lceil \log_2 k \rceil}$ to create the uniform superposition $|\Psi_0\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^k |b_i\rangle$.

Step 3: Parallel evaluation. By the linearity of unitary evolution, the circuit $U_{\mathcal{A}}$ that implements one step of $\mathcal{A}$ acts on all branches simultaneously: $U_{\mathcal{A}} |\Psi_0\rangle = \frac{1}{\sqrt{k}} \sum_i U_{\mathcal{A}} |b_i\rangle$. Since $\mathcal{A}$ runs in $O(n)$ steps, $U_{\mathcal{A}}$ consists of $O(n)$ layers, each implementable by a polynomial-size circuit.

Step 4: Amplitude amplification. Apply Grover-type amplitude amplification [1] to boost the amplitude of accepting branches. After $O(\sqrt{k})$ iterations, the probability of observing an accepting branch exceeds $\frac{2}{3}$.

Step 5: Terminal measurement. Measure the output qubit. This constitutes the single superposition decay event. By Theorem 5.1, the resulting problem is in BQP.

The total circuit size is $O(n \cdot \sqrt{k}) \leq O(n \cdot 2^{cn/2})$, which is polynomial for fixed $c < \frac{2 \log n}{n}$ — i.e., for linear $f(n) = cn$ with small c . This covers all algorithms with sub-exponential branch counts, which includes all deterministic and polynomially-branching non-deterministic linear-time algorithms. $\square$ $\square$

6.2 Linear Systems as Parallel DAGs

Theorem 6.2 (Linear Systems as Probabilistic DAGs). *Any system of m linear equations in n variables over $\mathbb{R}$ (or $\mathbb{C}$) with a unique solution can be expressed as a probabilistic DAG $G = (V, E, p)$ whose parallel evaluation under SD yields the solution in time $O(\log n)$ on a quantum system.*

Proof Sketch. The system $A\mathbf{x} = \mathbf{b}$ (with $A \in \mathbb{R}^{m \times n}$) has a unique solution $\mathbf{x}^* = A^{-1}\mathbf{b}$ (assuming A is invertible). The HHL algorithm [2] solves this system in $O(\log n \cdot \kappa^2)$ time on a quantum computer, where κ is the condition number of A . We model each variable as a DAG node and each dependency as a directed edge. Gaussian elimination induces a topological order on the DAG; each elimination step is a parallel unitary operation on the corresponding qubit register. The terminal measurement on each variable register constitutes a separate SD event, and the results compose to give $\mathbf{x}^*$. $\square$ $\square$

7 Rift: A Quantum-Classical Grammar Matcher

7.1 Overview

Rift (formally: *RIFT* — a *Flexible Translator*) is the compiler front-end component of the OBINexus toolchain. Its theoretical grounding in the SD framework arises from the observation that grammar matching in a parser shares the structural properties of quantum superposition: multiple production-rule paths may be simultaneously *active* (pending) until sufficient lookahead disambiguates them. The decay event corresponds to the moment the parser commits to a single parse path.

7.2 Wave Postulates as Triangular Matching Postulates

The wave postulate (Axiom 3) asserts that a quantum system in superposition $|\Psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ yields one of two outcomes upon measurement. We draw a direct structural analogy to a context-free grammar $G = (V, \Sigma, R, S)$ with two competing production rules $A \rightarrow \alpha \mid \beta$. In both cases:

- The system is in a *pending* superposition of both alternatives until a discriminating token (the “measurement”) is encountered.
- The collapse to a single rule (the “decay”) is the parser’s commit to one production.
- Destructive interference corresponds to the parser eliminating impossible parse paths via lookahead or semantic constraints.

Definition 7.1 (Grammar Superposition State). Let $G = (V, \Sigma, R, S)$ be a context-free grammar and let π be a partial parse of input $w = w_1 w_2 \cdots w_n$. The *grammar superposition state* at position i is the set

$$\Gamma_i = \{ (A \rightarrow \alpha \bullet \beta, j) \mid A \rightarrow \alpha\beta \in R, w_j \cdots w_i \Rightarrow^* \alpha \}$$

of all active Earley items. The state Γ_i is the grammar analogue of the superposition state vector: it encodes all simultaneously possible parse paths.

Definition 7.2 (Grammar Decay Event). A *grammar decay event* at position i occurs when $|\Gamma_i| = 1$, i.e., all but one parse path has been eliminated by lookahead. This corresponds to the collapse of the grammar superposition state to a single production, exactly as measurement collapses $|\Psi\rangle$ to a single basis state.

7.3 Parallel Grammar Evaluation

Theorem 7.3 (Rift Parallel Parsing). *Let G be an unambiguous context-free grammar and let w be an input string of length n . Rift evaluates all production rules of G in parallel by maintaining the full grammar superposition state Γ_i at each position, collapsing (decaying) only when the lookahead forces $|\Gamma_i| = 1$. The resulting parse is produced in $O(n^3)$ time in the worst case (Earley) and $O(n)$ for unambiguous grammars in Chomsky Normal Form, corresponding to a single-decay-event parse.*

Proof Sketch. The Earley algorithm maintains all active items in Γ_i at each position. For an unambiguous grammar, the number of active items is bounded by a constant at each position after the initial scan, so the grammar superposition state decays to a singleton in $O(n)$ steps. This is structurally identical to a BQP computation: the circuit evaluates all branches in parallel and the terminal measurement (grammar decay) yields the unique correct parse with probability 1. $\square$ $\square$

7.4 Stabilisation Principle

The SD framework introduces a *stabilisation principle* as the binding mechanism between the quantum measurement window and its cognitive or computational substrate.

Definition 7.4 (Stabilisation Principle). A system S obeys the *stabilisation principle* if, within the SD window, all intermediate states are reversible (unitary), and the terminal measurement commits the system to a stable classical output that cannot be further modified within the same decay window. Formally:

$$\forall t < t_{\text{decay}} : U(t) \text{ is unitary and reversible.} \quad \text{At } t_{\text{decay}} : \text{output is classical and irreversible.}$$

In the Rift context, this means: all grammar transitions before the decay event are reversible (backtracking is permitted); after the grammar decay event, the committed parse tree is final.

8 Quantum Burst Protocol (QBP)

8.1 Motivation

The *Quantum Burst Protocol* (QBP) extends the SD framework from computation to communication. Just as a quantum circuit collapses a superposition of computation branches to a single classical output, a QBP connection collapses a superposition of communication paths between two endpoints to a single established channel.

8.2 Formal Definition

Definition 8.1 (QBP Connection). A *Quantum Burst Protocol connection* is a tuple

$$\text{QBP}(A, B) = (G, \text{SD}(a, b), \Pi),$$

where:

- $G = (E, N)$ is the quantum connection graph (Definition 3.1), with $A, B \in N$ the sender and receiver endpoints.
- $\text{SD}(a, b)$ is the superposition decay window governing the connection establishment handshake.
- $\Pi = \{\pi_1, \pi_2, \dots, \pi_k\}$ is the set of candidate communication paths (edges in E) held in superposition during the handshake phase.

The connection is *established* (resolved) when the SD window closes and a single path $\pi^* \in \Pi$ is selected; it is *refused* (rejected) if no path survives the decay; and it is *negotiating* (pending) during the handshake phase.

8.3 QBP Handshake as Superposition Decay

Proposition 8.2 (QBP Three-Phase Handshake). *A QBP connection establishment proceeds in three phases corresponding exactly to the three states of the SD window:*

1. **Negotiating (pending):** Both A and B maintain a superposition of all feasible communication paths Π . No classical commitment is made.
2. **Resolved:** The SD window closes; the path π^* with maximal probability amplitude is selected. A and B are classically connected via π^* .
3. **Rejected:** The SD window closes with all path amplitudes below the threshold $b(n)$; no connection is established.

Remark 8.3. The QBP handshake is a quantum analogue of the classical TCP three-way handshake (SYN / SYN-ACK / ACK), with the key difference that in QBP, *all candidate paths are simultaneously active* during the negotiating phase, and the path selection is determined by a single quantum measurement (the decay event) rather than sequential message exchange.

8.4 Spectral Graph Representation

The candidate path set Π is naturally represented as a *spectral graph*:

Definition 8.4 (Spectral Connection Graph). The *spectral connection graph* of a QBP connection is the directed graph $\hat{G} = (V, E, \lambda)$ where $\lambda : E \rightarrow [0, 1]$ assigns each edge a probability amplitude (eigenvalue weight) such that $\sum_{e \in \Pi} \lambda(e)^2 = 1$. The selected path π^* is the eigenvector corresponding to the dominant eigenvalue after the decay measurement.

Example 8.5 (Internet Connection as QBP). Consider connecting a client A to a server B over a network with k possible routing paths. In classical networking, these paths are evaluated sequentially (or via parallel probes with separate classical outcomes). Under QBP, the network interface holds all k paths in superposition:

$$|\Pi\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^k |\pi_i\rangle.$$

Quality-of-service metrics (latency, bandwidth, packet loss) act as the observable $\hat{O}$. Measurement of $\hat{O}$ collapses $|\Pi\rangle$ to the optimal path $|\pi^*\rangle$ in a single decay event, achieving optimal routing in $O(1)$ measurement time (assuming the superposition can be prepared in $O(\log k)$ time).

9 MMUKO OS and the OBINexus Toolchain

9.1 MMUKO OS

MMUKO OS (M for Mike, Q for Bella [Queen], K for Kilo, O for Oscar) is the operating system substrate for the OBINexus quantum-classical computing framework. The name encodes its philosophical foundation: *“the spirit of good and evil that connects nothing and everything, that binds all of us and all of them.”* This duality—nothing and everything, good and evil—maps directly onto the binary superposition state $|0\rangle$ and $|1\rangle$ and their linear combinations.

MMUKO OS is a cloud-based Windows system that serves as the classical substrate on which quantum algorithms are emulated prior to the availability of dedicated quantum hardware. Its design is guided by three principles:

1. **Read-Write symmetry:** All memory operations respect the RWX model (Section 4), ensuring that write-heavy workloads collapse atomically to coherent read outcomes.
2. **Superposition-aware scheduling:** The OS scheduler treats parallel threads as superposed branches of a single computation, deferring classical commitment until the SD window closes.
3. **Decay-triggered synchronisation:** Inter-process communication is governed by QBP handshakes (Section 8), replacing traditional lock-based synchronisation with measurement-based collapse events.

9.2 Toolchain Stack

The OBINexus toolchain implements the SD framework across the full compilation pipeline:

Component	Role in SD Framework
<code>riftlang.exe</code>	Source language compiler; implements grammar superposition (Section 7) to parse Rift source files in parallel.
<code>.so.a</code>	Shared object / static archive; intermediate representation holding the superposed (pre-decay) compilation state.
<code>rift.exe</code>	Linked executable; represents the post-decay committed classical binary output.
<code>gosilang</code>	Target runtime language binding; maps SD primitives to hardware-level instructions.
<code>nlink</code>	Build linker / orchestrator; manages the RWX transaction across compilation units.
<code>polybuild</code>	Polyglot build system; coordinates multiple SD decay events across heterogeneous language targets.

The pipeline can be summarised as:

`riftlang.exe` $\rightarrow$ `.so.a` $\rightarrow$ `rift.exe` $\rightarrow$ `gosilang` $\xrightarrow{\text{nlink / polybuild}}$ deployed binary.

10 Conclusion and Future Work

We have formalised the concept of *Superposition Decay* as a measurement window bridging quantum and classical computation. The main results are:

- The SD window is a three-valued construct (resolved / rejected / pending) that generalises the binary output of a classical TM and aligns precisely with the promise-problem formulation of quantum complexity classes.
- The RWX memory model provides an operational interpretation of the decay window: n write operations and k read operations within the window collapse atomically to a single coherent read.
- BQP is exactly the class of promise problems solvable within a single SD event with a probability gap of $\frac{1}{3}$.
- Classical algorithms bounded by linear time complexity—deterministic or non-deterministic—can be simulated in parallel within a single SD window by a polynomial-size quantum circuit.
- Systems of linear equations can be expressed as probabilistic DAGs whose parallel evaluation under SD yields solutions in $O(\log n)$ quantum time.
- Rift’s parser exploits grammar superposition to evaluate all production rules in parallel, collapsing to a unique parse tree at a grammar decay event governed by the stabilisation principle.
- The Quantum Burst Protocol (QBP) extends SD to networking: all candidate communication paths are held in superposition during handshake, with optimal path selection achieved by a single spectral decay measurement.
- The MMUKO OS / OBINexus toolchain implements all of the above across the full pipeline: `rifklang.exe` $\rightarrow$ `.so.a` $\rightarrow$ `rift.exe` $\rightarrow$ `gosilang`, orchestrated by `nlink` and `polybuild`.

10.1 Future Work

The following directions are planned within the OBINexus research programme:

1. **Hardware realisation.** Implementing the SD framework on near-term quantum hardware (superconducting qubits, trapped ions) via the MMUKO OS / `rifklang` compiler toolchain.
2. **Multi-decay protocols for PP and PSPACE.** Designing iterated SD protocols that amplify probability across multiple decay events to address problems beyond BQP.
3. **QBP networking stack.** Implementing QBP as a full networking protocol atop the MMUKO OS kernel, replacing TCP handshakes with spectral decay measurements for latency-optimal routing.
4. **Rift compiler validation.** Formal verification that the Rift grammar decay event produces parse trees equivalent to those of a classical Earley parser, with empirical benchmarks on `gosilang` target programs.
5. **Cognitive binding.** Exploring the stabilisation principle as a model for attention and commitment in cognitive architectures: the SD window as a formal model for the moment of decision in neural-symbolic systems.

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